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Measurement of phase equilibria of supercritical carbon dioxide and paraffins

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Abstract

High pressure phase equilibria of supercritical carbon dioxide with *n*-alkanes, *n*-C_{12}, *n*-C_{16}, *n*-C_{20}, *n*-C_{24}, *n*-C_{28} and *n*-C_{36}, have been measured over the temperature range of 313–367 K, and are reported in this work. Of the various equations of state investigated, it was found that the Patel Teja equation of state provided the best fit of the CO_2–*n*-alkane systems. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Supercritical; Phase equilibria; *n*-Alkanes; Carbon dioxide; Equations of State

1. Introduction

It is believed that supercritical extraction is a viable alternative to short path distillation, liquid–liquid extraction and crystallisation, as a means of producing high molecular weight alkane fractions with low polydispersities [1,2]. The ease with which the solubilities in supercritical fluids can be manipulated as well as the low operating temperatures required for supercritical extraction make it attractive for the fractionation of heat labile hydrocarbons. Carbon dioxide is readily available, cheap, non-toxic and non-flammable, making it a favoured solvent for use in the extraction process. In order to study the supercritical fluid fractionation process, a reliable equation of state is needed to be able to model the process. Reliable dew point data is of paramount importance in determining the interaction parameters of the equation of state. There is currently however, a very limited number of datasets available on the phase equilibria of supercritical solvents and high molecular weight *n*-alkanes with datapoints on the dew point curve [3]. Although renowned experimentalists have reported datasets for CO_2 and alkanes [4–10,14–17], these datasets do not include enough dew point measurements to facilitate an accurate study of the possibilities of supercritical fluid fractionation. It should also be mentioned that there are several CO_2–alkane datasets of questionable accuracy in the literature. The CO_2–*n*-C_{16} datasets published by D'Souza et al. [11] and Charoensombut-Amon et al. [12] follow completely different trends, and it is impossible to determine which is the better set of data

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without doing further experimental work. Gasem and Robinson [6] pointed out the same type of discrepancy. Dew point data sets have been reported by the group of Schneider [8–10,16] but the *n*-alkane mass fractions were only determined to an accuracy of 1.5 mass % [16].

The aim of this work is to measure accurate phase equilibrium data for CO_2–alkane systems that can be used in the study of supercritical fluid fractionation processes, and entails phase equilibria measurements at temperatures just above the melting points of the alkanes with special emphasis on measurements along the dew point curve.

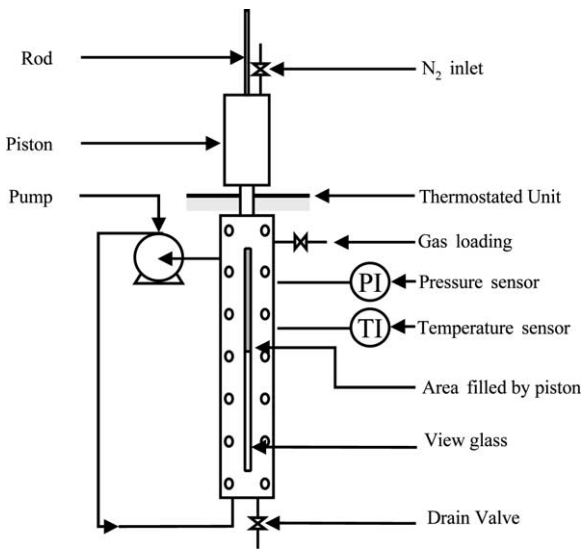


Fig. 1. View cell used to study the CO_2–*n*-C_{12} and CO_2–*n*-C_{16} systems.

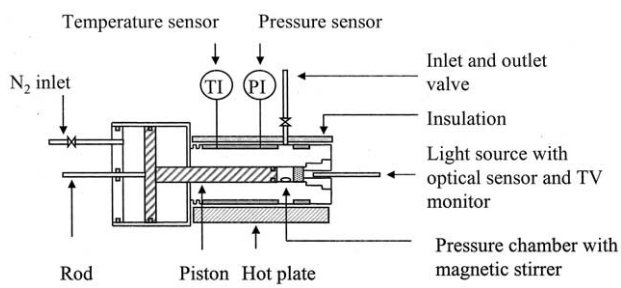


Fig. 2. View cell used to study the CO_2–*n*-C_{20}, *n*-C_{24}, *n*-C_{28} and *n*-C_{36} systems.

2. Experimental set-up

An indirect method of phase equilibrium measurement was used to study the binary phase equilibria.

The experimental set-up as discussed by du Rand [3,13] was used to measure the phase equilibria of the CO_2–*n*-C_{12} and *n*-C_{16} systems. A smaller view cell with a higher operating pressure was used to measure the phase equilibria of the CO_2 with *n*-C_{20}, *n*-C_{24}, *n*-C_{28} and *n*-C_{36}. See Figs. 1 and 2 for the experimental set-up used.

Both phase equilibrium cells operate on the principle that, under isothermal conditions, the pressure of the cell is changed by changing the volume. Upon changing the pressure, the phase boundary will be crossed and this phase transition is detected visually. The composition of the primary phase is taken to be that of the total composition loaded into the cell.

3. Experimental procedure

The same experimental technique is used on the two different phase equilibrium cells, and is discussed in detail by du Rand [13]. Only a brief description of the method is given here:

A known mass of the *n*-alkane is loaded into the cell chamber. Once the alkane is loaded into the cell, all the air is removed from the system by evacuating the cell to a pressure of 80 Pa using a multistage vacuum pump. The cell is then flushed several times with the solvent at low pressures in order to remove any final traces of air from the system.

The cell is loaded with CO_2 by using a gas bomb filled with liquid CO_2. Prior to loading the bomb with liquid CO_2, it is flushed several times with CO_2 to remove all air from the bomb-chamber. The loaded bomb is connected to the cell at the gas loading port, and the connecting line (20 mm long, 4 mm ID) between the bomb and the cell evacuated in order to remove the air from the lines. The solvent that remained in the cell as a result of the flushing is released through the con-

Table 1
Materials used in the experiments

Component	Source	Purity (%)	T_{MELT} (K)
Dodecane n-C_{12}	Aldrich	>99	264
Hexadecane n-C_{16}	Aldrich	99	291*
Eicosane n-C_{20}	Sigma	99	310
Tetracosane n-C_{24}	Sigma	≈99	322–324*
	Fluka	>99	
Octacosane n-C_{28}	Fluka	≥98	330–335*
Hexatriaconane n-C_{36}	Fluka	98	349
	Humphry Chemical Company	99	

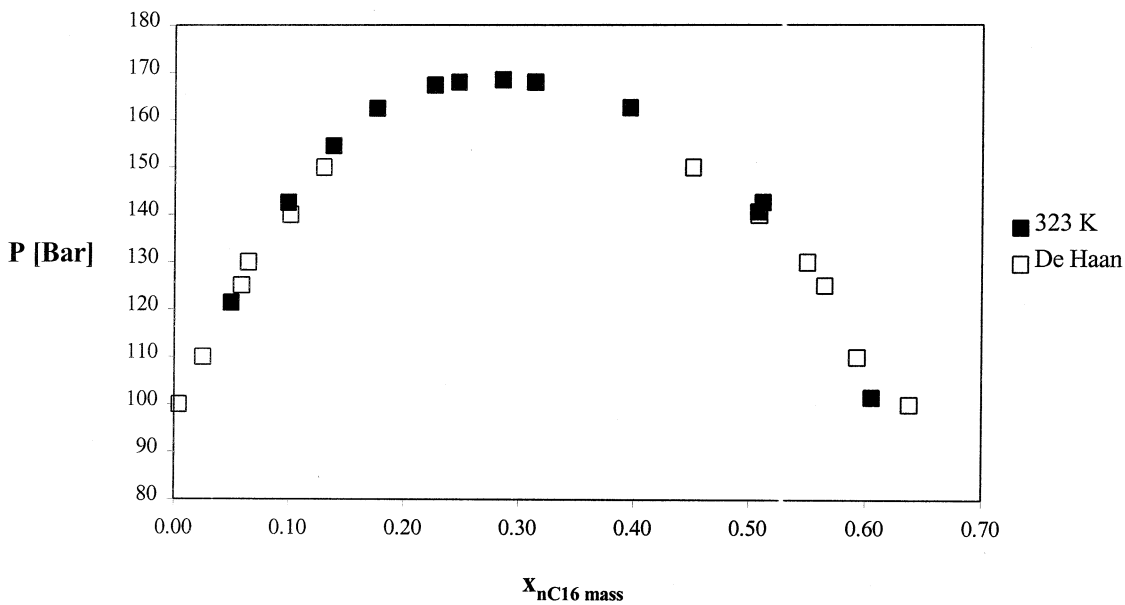


Fig. 3. Comparison between experimentally determined data and data measured by De Haan [14] of the system CO_2–n-C_{16}.

necting line to remove the last traces of air from the system. The solvent is forced into the cell by heating the bomb to a temperature higher than that of the cell chamber. The valve on the bomb is closed and the connecting line is heated to ensure that only CO_2 gas is left in it. Approximately, 0.02 g of CO_2 is lost from the connecting line. The bomb is weighed before and after loading of the cell to determine the amount of CO_2 added. Both the gas bomb and the amount of solute are weighed to an accuracy of 0.001 g.

After loading, the cell is heated to the desired temperature. Once the alkane has melted, the system pressure is increased to force the cell contents into a single phase. This will ensure that the composition in the cell remains constant even if there are minute leaks in the system.

When thermal equilibrium in the one-phase region has been reached, the pressure in the cell is gradually reduced until the formation of a second phase is observed. The phase boundary is located in the pressure interval between the points where it is possible to distinguish between a single- or a two-phase state. The size of this pressure interval is reduced by alternately increasing and decreasing the pressure a number of times, each time observing the phase changes in the cell. The technique ensures that the measurement of metastable

Table 2
CO_2–n-C_{12} phase equilibrium data

CO_2–n-C_{12} system							
Temperature 313.2 K		Temperature 323.2 K		Temperature 333.2 K		Temperature 343.2 K	
P_{abs} (bar)	$z_{mass\ n\text{-}C_{12}}$	P_{abs} (bar)	$z_{mass\ n\text{-}C_{12}}$	P_{abs} (bar)	$z_{mass\ n\text{-}C_{12}}$	P_{abs} (bar)	z_{mass} n-C_{12}
85.7	0.028	99.0	0.028	110.4	0.028	120.0	0.028
84.9	0.051	101.3	0.051	117.5	0.051	130.5	0.051
84.2	0.103	102.8	0.103	122.1	0.103	141.0	0.103
83.5	0.200	102.9	0.200	123.3	0.200	143.0	0.200
82.8	0.317	101.7	0.317	121.6	0.317	140.0	0.317
82.2	0.405	99.1	0.405	117.3	0.405	134.3	0.405
79.5	0.483	94.4	0.483				

equilibria is avoided. The combined inaccuracy in the phase boundary pressures as a result of the pressure transmitter and the optical technique for determining the phase boundary, is estimated to be less than 1 bar.

The composition of the primary phase is taken to be that of the total composition loaded into the cell. The piston displacement is also measured at this point and the cell volume and system density calculated.

4. Materials

The materials used in the experiments are listed in Table 1. These chemicals were used without any further purification. However, it should be noted that during the loading process the alkanes were degassed under vacuum conditions for up to 30 min.

Industrial grade nitrogen was used in the pressure control.

5. Comparison with published data

Fig. 3 shows good agreement between data previously published by de Haan [14] for the system CO_2–n-C_{16} at 323 K and the same data set measured on the cell as shown in Fig. 1.

The degree of correlation between the new experimental data and the published results serves as a verification of the experimental techniques used in this work.

6. Experimental results

Tables 2–7 contain the experimentally determined CO_2–n-alkane phase equilibria and densities at the phase boundaries.

Table 3
CO_2–n-C_{16} phase equilibrium data

CO_2–n-C_{16} system			
Temperature 313.2 K		Temperature 323.2 K	
P_{abs} (bar)	$z_{mass\ n\text{-}C_{16}}$	P_{abs} (bar)	z_{mass} n-C_{16}
99.0	0.050	121.4	0.050
123.9	0.099	142.5	0.099
142.0	0.138	154.5	0.138
154.7	0.176	162.4	0.176
162.8	0.226	167.3	0.226
163.7	0.247	167.9	0.247
165.1	0.313	168.4	0.285
159.8	0.396	168.0	0.313
146.6	0.442	162.6	0.396
132.0	0.507	140.5	0.507
133.8	0.511	142.6	0.511
111.1	0.529	101.5	0.606
95.0	0.588		
85.5	0.606		

Table 4
CO_2–n-C_{20} phase equilibrium data

CO_2–n-C_{20} system														
Temperature 315.9 K			Temperature 320.6 K			Temperature 329.6 K			Temperature 339.1 K			Temperature 348.3 K		
P_{abs} (bar)	z_{mass} n-C_{20}	ρ (g/cm^3)	P_{abs} (bar)	z_{mass} n-C_{20}	ρ (g/cm^3)	P_{abs} (bar)	z_{mass} n-C_{20}	ρ (g/cm^3)	P_{abs} (bar)	z_{mass} n-C_{20}	ρ (g/cm^3)	P_{abs} (bar)	z_{mass} n-C_{20}	ρ (g/cm^3)
100.9	0.018	0.778	109.1	0.018	0.739	121.8	0.018	0.657	135.0	0.018	0.616	140.9	0.018	0.552
196.1	0.055	1.167	157.4	0.034	1.169	166.2	0.034	1.098	186.8	0.034	1.061	203.1	0.034	1.015
249.6	0.502	1.196	197.9	0.055	1.125	205.0	0.055	1.063	212.3	0.055	1.006	219.7	0.055	0.948
185.2	0.559	1.102	223.1	0.502	1.172	258.9	0.470	1.094	281.4	0.132	1.168	279.0	0.132	1.104
106.0	0.619	1.091	175.6	0.559	1.088	205.6	0.502	1.117	303.5	0.224	1.139	293.2	0.224	1.084
77.4	0.660	0.942	110.3	0.619	1.083	175.3	0.559	1.069	300.0	0.277	1.217	291.2	0.277	1.168
			86.4	0.660	0.989	118.6	0.619	1.079	297.1	0.359	1.059	287.9	0.359	1.030
						104.3	0.660	1.019	246.5	0.470	1.052	247.9	0.470	1.024
									203.3	0.502	1.121	188.6	0.559	1.032
									180.3	0.559	1.049	138.5	0.619	1.054
									129.4	0.619	1.066	115.0	0.660	0.933
									104.3	0.660	0.936			

Table 5
CO_2–n-C_{24} phase equilibrium data

CO_2–n-C_{24} system											
Temperature 329.7 K			Temperature 339.2 K			Temperature 348.1 K			Temperature 357.1 K		
P_{abs} (bar)	$z_{mass\ n\text{-}C_{24}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{24}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{24}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{24}}$	ρ (g/cm^3)
190.4	0.022	0.921	203.6	0.022	0.875	213.8	0.022	0.826	226.6	0.022	0.791
246.2	0.035	1.097	234.5	0.035	1.012	249.7	0.035	0.971	279.8	0.046	1.003
281.4	0.046	1.199	283.7	0.046	1.165	279.8	0.046	1.073	301.6	0.507	1.062
258.9	0.561	1.102	239.6	0.561	1.082	235.1	0.561	1.065	239.5	0.561	1.043
216.0	0.583		208.9	0.583		212.0	0.583		217.8	0.583	
174.9	0.618		176.8	0.618		184.3	0.618		192.4	0.618	
154.8	0.627	1.055	166.0	0.627	1.047	172.4	0.627	1.022	185.0	0.627	

Table 6
CO_2–n-C_{28} phase equilibrium data

CO_2–n-C_{28} system											
Temperature 338.5 K			Temperature 348.0 K			Temperature 357.0 K			Temperature 366.4 K		
P_{abs} (bar)	$z_{mass\ n\text{-}C_{28}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{28}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{28}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{28}}$	ρ (g/cm^3)
196.8	0.017	0.791	226.3	0.017	0.785	239.8	0.017	0.760	254.8	0.019	0.751
232.5	0.019	0.019	232.4	0.019	0.821	242.2	0.019	0.783	274.8	0.026	0.856
246.6	0.026	0.026	276.2	0.026	0.983	272.2	0.026	0.915	288.8	0.028	0.910
278.7	0.028	0.028	289.9	0.028	1.028	284.4	0.028	0.961	289.5	0.569	
272.9	0.617	0.617	257.3	0.617	1.093	282.5	0.569		253.2	0.617	1.061
272.6	0.622	0.622	257.1	0.622	1.092	252.7	0.617	1.081	254.5	0.622	1.052
200.9	0.632	0.632	199.1	0.632		252.1	0.622	1.074	212.9	0.632	
140.1	0.685	0.685	148.4	0.685	0.941	207.4	0.632		172.3	0.685	0.920
						162.3	0.685	0.937			

Table 7
CO_2–n-C_{36} phase equilibrium data

CO_2–n-C_{36} system											
Temperature 344.0 K			Temperature 3448.5 K			Temperature 357.0 K			Temperature 366.9 K		
P_{abs} (bar)	$z_{mass\ n\text{-}C_{36}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{36}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{36}}$	ρ (g/cm^3)	P_{abs} (bar)	$z_{mass\ n\text{-}C_{36}}$	ρ (g/cm^3)
289.6	0.008	0.923	285.4	0.008	0.849	289.8	0.008	0.764	289.7	0.008	0.799
223.2	0.014	0.825	254.4	0.010	0.832	260.6	0.010	0.730	260.0	0.010	0.760
258.2	0.679	1.108	234.3	0.014	0.792	250.0	0.014	0.710	254.4	0.014	0.741
237.4	0.693	1.097	263.8	0.019	0.893	266.3	0.019	0.796	272.6	0.019	0.823
201.0	0.713	1.075	292.6	0.022	1.043	276.9	0.022	0.859	273.1	0.022	0.883
			297.5	0.660	1.073	291.3	0.660	1.032	285.6	0.660	
			254.7	0.679	1.083	250.7	0.679	1.047	246.8	0.679	1.064
			233.7	0.693	1.069	231.5	0.693	1.044	235.1	0.693	1.061
			201.4	0.713	1.050	205.3	0.713	1.025	211.5	0.713	1.040

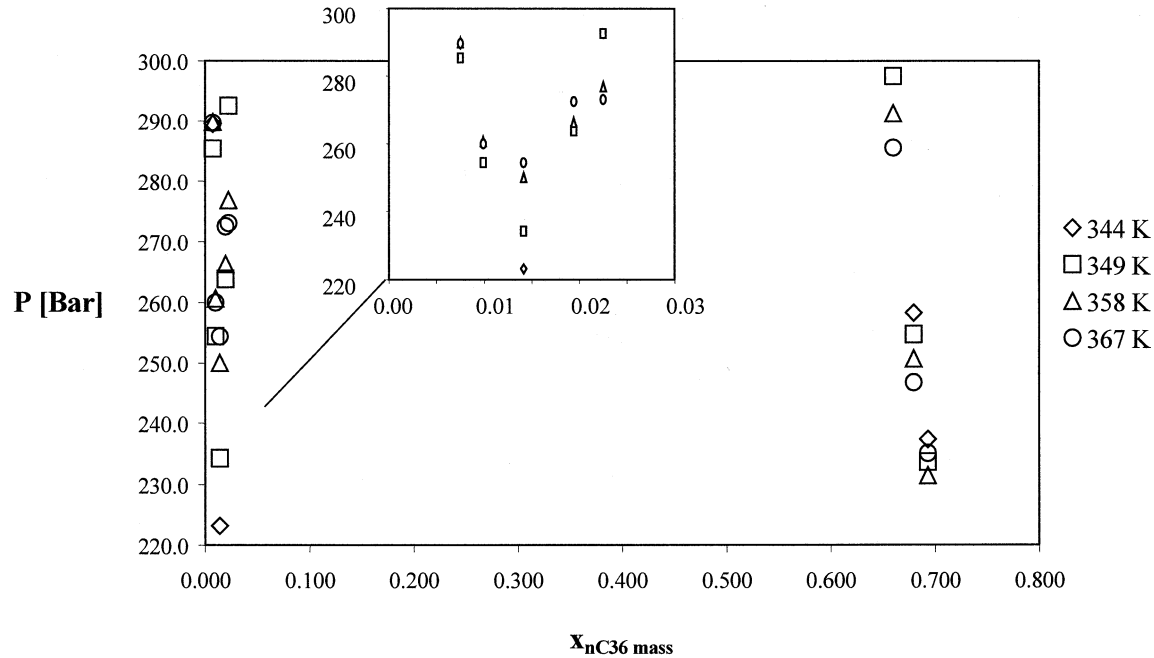


Fig. 4. P–x diagram of the CO_2–n-C_{36} system.

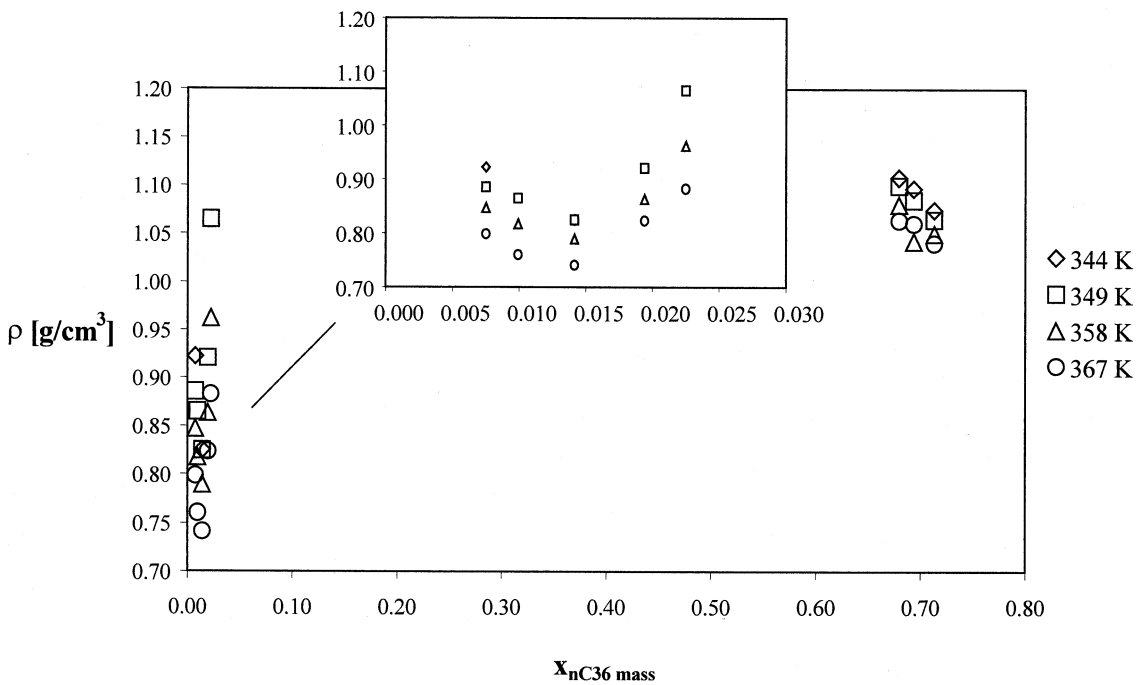


Fig. 5. ρ–x diagram of the CO_2–n-C_{36} system.

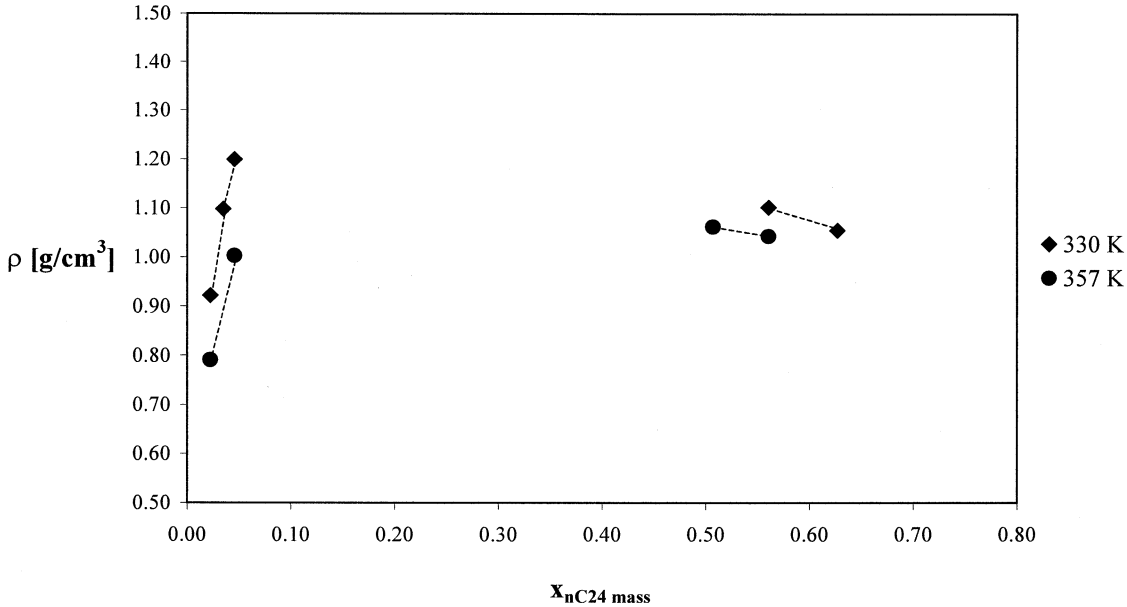


Fig. 6. ρ–x diagram of the CO_2–n-C_{24} system.

It was observed that at very low n-alkane concentrations the CO_2–n-C_{36} system exhibits a decrease in the phase boundary pressure up to an n-alkane mass fraction of 0.014. At concentrations lower than this value, an increasingly higher pressure was needed to force the components into a single phase. This indicates the existence of another liquid–vapour dome at low n-C_{36} concentrations. The CO_2–n-C_{36} mixture density plot also follows a similar trend. The trends are shown in Figs. 4 and 5. It has however been indicated by Fall et al. [17], that for CO_2–n-alkane systems displaying type III phase behaviour, the LLV locus is obscured by the solidification of the alkane in systems with alkanes longer than n-C_{22}. It is not known whether the other CO_2–n-alkane systems in this study also exhibit the formation of an additional liquid–vapour dome at very low n-alkane concentrations, since no measurements were made at these low n-alkane concentrations. It is an area worthy of further investigation.

All the CO_2–n-alkane systems in this study exhibit a solvent and solute phase density inversion. This has the effect that at certain concentrations and pressures the CO_2 rich phase will have a higher density than the alkane rich phase. Fig. 6, a plot of the CO_2–n-C_{24} system densities, clearly illustrates the inversion phenomena. This density behaviour has to be borne in mind during the operation of supercritical fluid paraffin fractionation columns where CO_2 is used as the solvent.

7. Equations of state

Four equations of state, EOS, were used to model the experimental data. The Van der Waals type equations of state: the Soave–Redlich–Kwong EOS (SRK) as reported by Soave [18]; the modified Peng Robinson EOS (PR) developed by Stryjek and Vera [19,20] as used by Nieuwoudt

Table 8
Pure component parameters

Component	T_c (K)	P_c (bar)	ω
CO_2	304.2	73.83	0.224
n-C_{12}	658.0	18.20	0.576
n-C_{16}	720.6	14.19	0.744
n-C_{20}	768.0	11.60	0.907
n-C_{24}	804.0	9.80	1.071
n-C_{28}	832.0	8.50	1.238
n-C_{36}	874.0	6.80	1.526

Table 9
Interaction parameters determined for the CO_2–*n*-alkane systems

n-Alkane	*T* (K)	SRK		PR		PT	
		k_{ij}	l_{ij}	k_{ij}	l_{ij}	k_{ij}	l_{ij}
n-C_{12}	343	0.0900	0.0350	0.0870	0.0350	0.0543	0.0781
	333	0.0900	0.0300	0.0820	0.0350	0.0530	0.0650
	323	0.0940	0.0150	0.0820	0.0350	0.0530	0.0650
	313	0.0976	0.0269	0.0820	0.0390	0.0600	0.0550
n-C_{16}	323	0.0940	0.0300	0.0840	0.0250	0.0464	0.0738
	313	0.0945	0.0700	0.0815	0.0250	0.0480	0.0700
n-C_{20}	348	0.0979	0.0720	0.0798	0.1195	0.0317	0.195
	339	0.0979	0.0562	0.0820	0.1200	0.0350	0.180
	330	0.1040	0.0646	0.0922	0.0720	0.0464	0.151
	321	0.1050	0.0600	0.0931	0.0686	0.0410	0.098
	314	0.1000	0.0550	0.0931	0.0686	0.0470	0.100
n-C_{24}	357	0.102	0.0766	0.0899	0.0925	0.0296	0.225
	348	0.102	0.0694	0.0895	0.0832	0.0301	0.194
	339	0.097	0.0375	0.0858	0.0585	0.0345	0.181
	330	0.101	0.0494	0.0852	0.0499	0.0310	0.101
n-C_{28}	366	0.0968	0.0598	0.0830	0.0900	0.0225	0.295
	357	0.0981	0.0557	0.0830	0.0800	0.0284	0.278
	348	0.0981	0.0557	0.0830	0.0800	0.0250	0.190
	339	0.0981	0.0557	0.0810	0.0500	0.0210	0.101
n-C_{36}	367	0.0858	0.0185	0.0700	0.0400	0.0073	0.210
	358	0.0898	0.0270	0.0700	0.0300	0.0137	0.207
	349	0.0799	−0.0021	0.0682	0.0222	0.0110	0.166
	344	0.0753	−0.0130	0.0652	0.0137	0.0056	0.132

[21]; the Patel–Teja EOS (PT) developed by Patel and Teja [22], and a Statistical Mechanical equation of state: the Cubic Simplified Perturbed Hard Chain equation of state (CSPHC) as reported by Wang and Guo [23]. The pure component parameters used in these equations of state are listed in Table 8. The performance of the equations (in the forms that they were published) was assessed and no attempts were made to modify the equations.

The following mixing and combining rules using two interaction parameters, k_{ij} and l_{ij}, were used with the Van der Waals type equations of state:

$$a = \sum_i \sum_j x_i x_j a_{ij}, \tag{1}$$

$$b = \sum_i \sum_j x_i x_j b_{ij}. \tag{2}$$

And for the case of the Patel Teja Equation with an additional *c* parameter:

$$c = \sum_i x_i c_i. \tag{3}$$

In these equations, *x* denotes the component mole fraction; *a*, *b* and *c* the various EOS parameters and a_{ii}, b_{ii} and c_i the pure component parameters of component *i*. The cross parameters, a_{ij} and b_{ij}, are determined by the following combining rules:

$$a_{ij} = (1 - k_{ij})\sqrt{a_i a_j}, \tag{4}$$

$$b_{ij} = (1 - l_{ij})\left(\frac{b_i + b_j}{2}\right). \tag{5}$$

Similar mixing and combining rules as the rules used by Wang and Guo [23] were used to extend the CSPHC equation of state for use with mixtures. Two interaction parameters, one for the

segment interaction energy and one for the segment size, were used.

The binary parameters of the Van der Waals type equations of state were fitted to the experimental data by using a computer package developed by Crause [24]. The fitting was done by repeatedly solving the flash algorithm with the various equations of state at specific phase

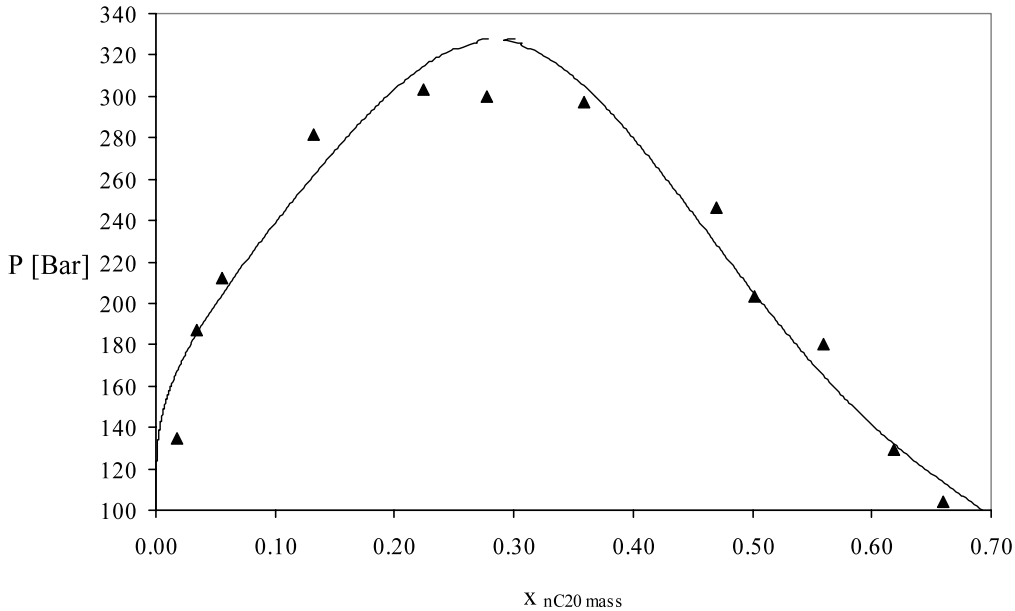


Fig. 7. Attempts to model the CO_2–n-C_{20} system at 348 K with the CSPHC EOS.

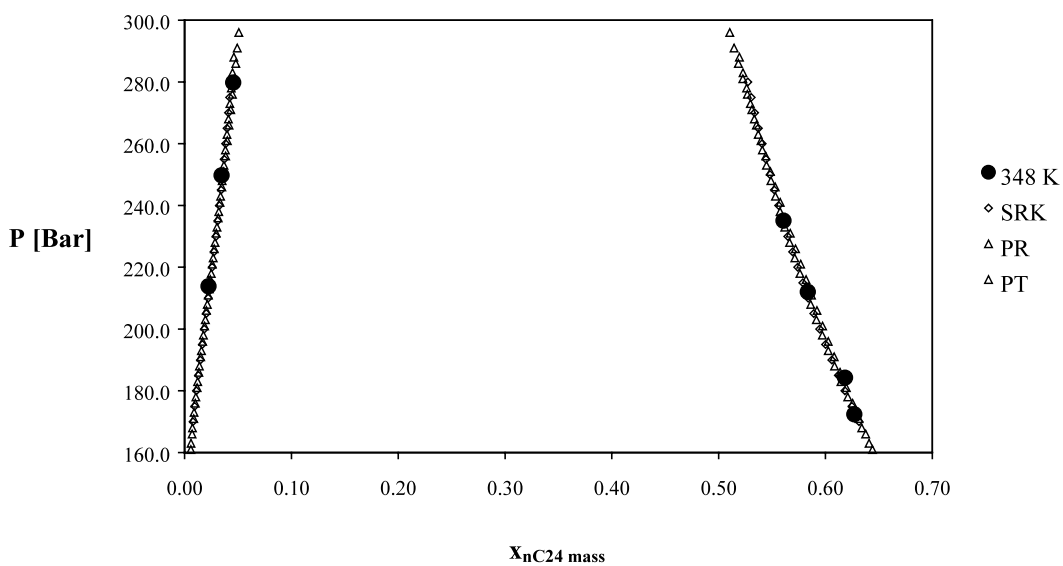


Fig. 8. P–x diagram of the CO_2–n-C_{24} system at 348 K with SRK k_{ij} 0.102, l_{ij} 0.0694; PR k_{ij} 0.0895, l_{ij} 0.0832; PT k_{ij} 0.0301, l_{ij} 0.194.

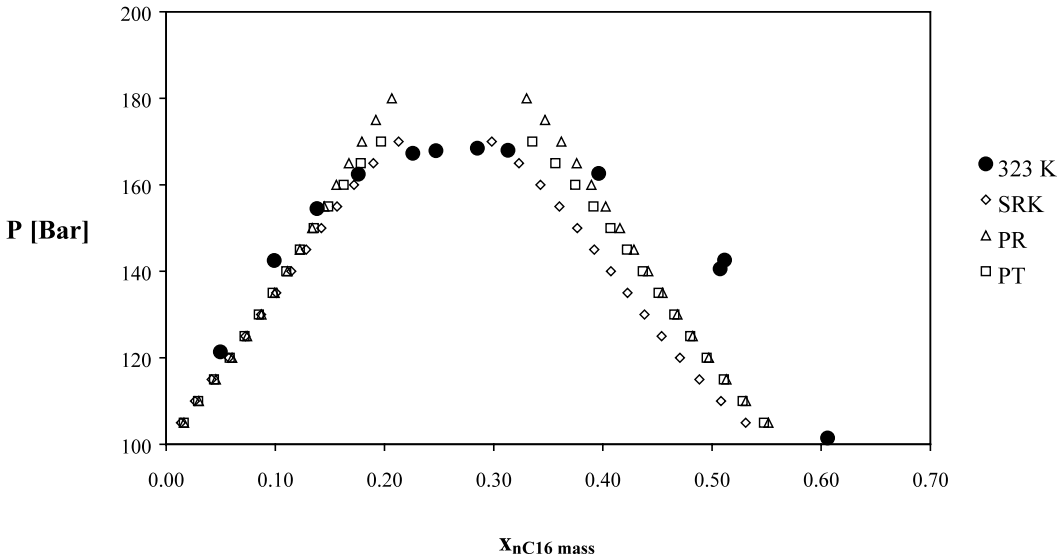


Fig. 9. P–x diagram of the CO_2–n-C_{16} system at 323 K with SRK k_{ij} 0.094, l_{ij} 0.03; PR k_{ij} 0.084 l_{ij} 0.025; PT k_{ij} 0.0464, l_{ij} 0.0738.

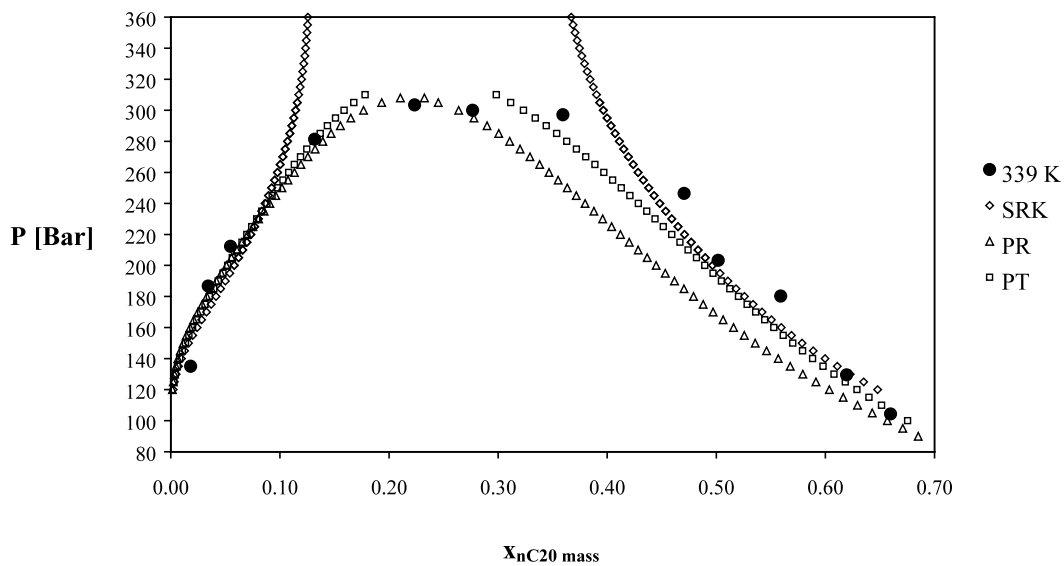


Fig. 10. P–x diagram of the CO_2–n-C_{20} system at 339 K with SRK k_{ij} 0.0979, l_{ij} 0.0562; PR k_{ij} 0.082, l_{ij} 0.12; PT k_{ij} 0.035, l_{ij} 0.18.

boundary pressures. The equations of state were forced to fit the data by adjusting the binary interaction parameters in order to reduce the observed deviations until an optimal fit could be achieved. The CSPHC equation of state was fitted similarly, using computer program developed by Nieuwoudt [21] to execute the flash algorithm at the various system pressures.

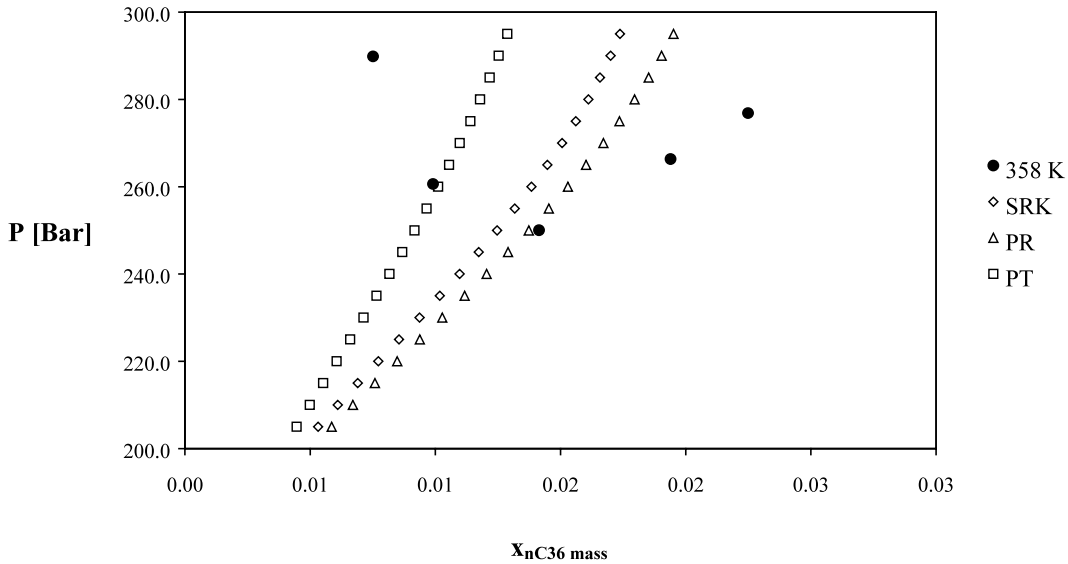


Fig. 11. P–x diagram of the CO_2–n-C_{36} system at 358 K with SRK k_{ij} 0.0898, l_{ij} 0.0270; PR k_{ij} 0.07, l_{ij} 0.03; PT k_{ij} 0.0137, l_{ij} 0.207.

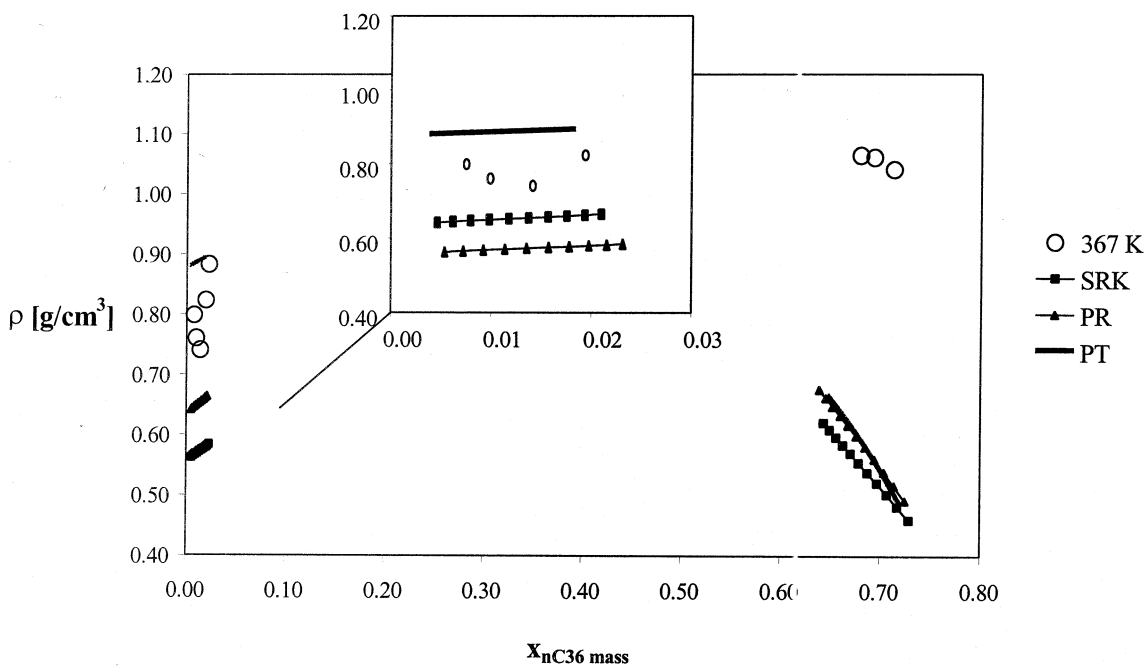


Fig. 12. ρ–x diagram of the CO_2–n-C_{36} system at 367 K with SRK k_{ij} 0.0858, l_{ij} 0.0185; PR k_{ij} 0.07, l_{ij} 0.04; PT k_{ij} 0.0073, l_{ij} 0.21.

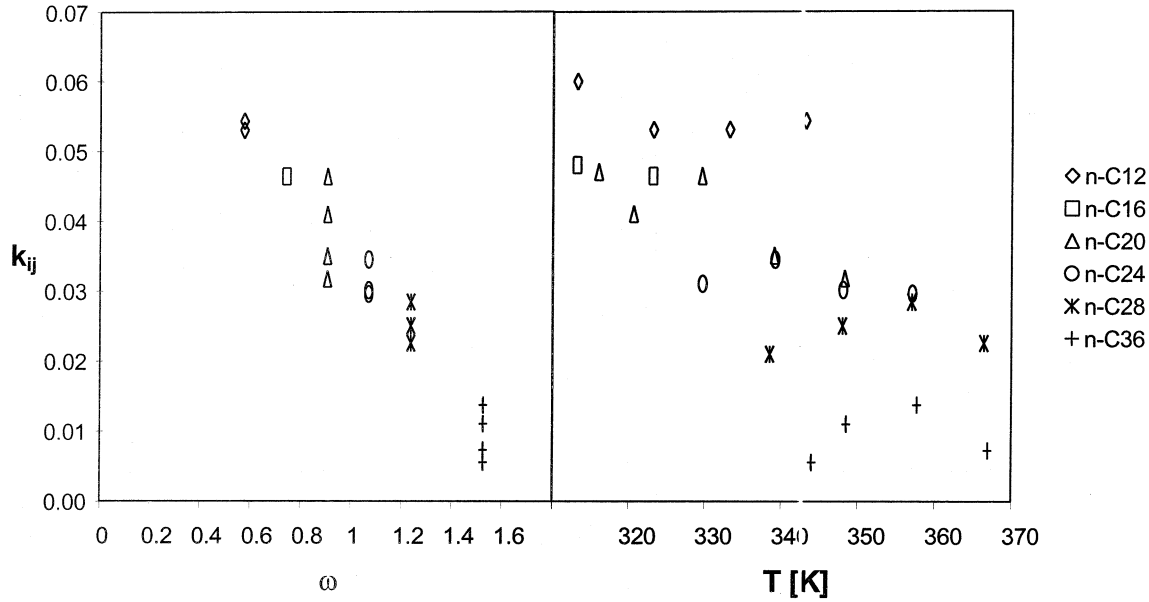


Fig. 13. Plot of the PT k_{ij} interaction parameters against the n-alkane acentric factor and the system temperature.

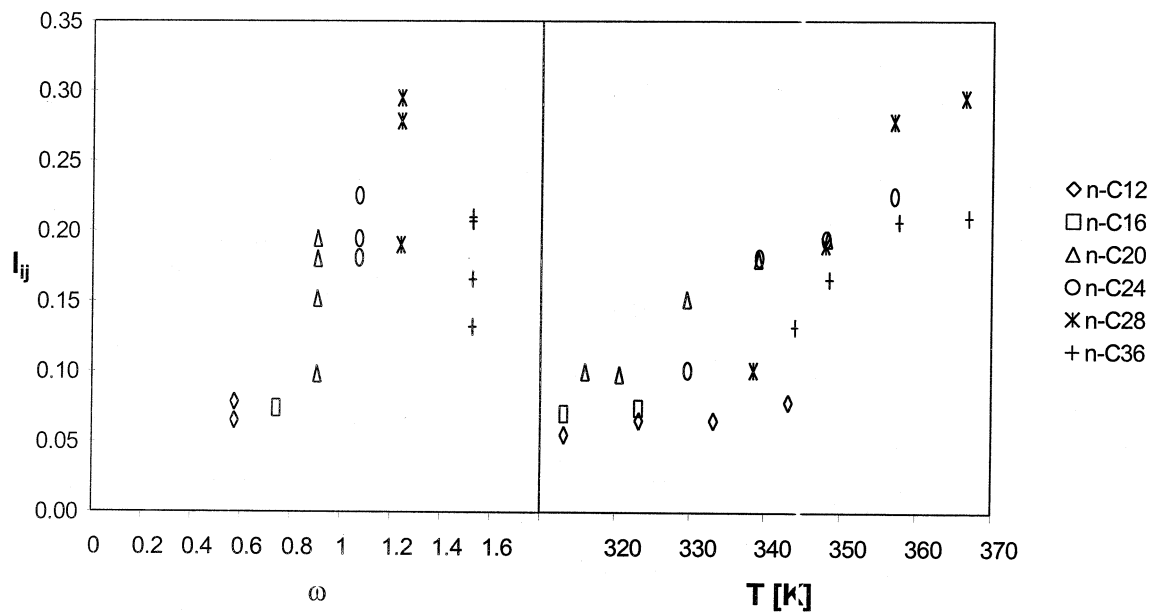


Fig. 14. Plot of the PT l_{ij} interaction parameters against the n-alkane acentric factor and the system temperature.

8. Modelling results

The binary interaction parameters determined for the SRK, PT and PR equations of state fitted to the CO_2–n-alkane data are represented in Table 9. In comparing Figs. 7 and 10 it is evident that the CSPHC EOS does not offer any real advantages over the Patel Teja or Peng Robinson

equations of state for the systems in this study. Although the CSPHC interaction parameters are smaller, the fit of the dew point line is not as good.

It is relatively easy to fit the interaction parameters of the equations of state to the experimental data in the regions far away from the mixture critical point. The phase boundaries of the systems CO_2–n-C_{24} to CO_2–n-C_{36} were measured at pressures far below the mixture critical pressure, and it can be seen on Fig. 8 that at these conditions all of the EOS provide a virtually identical fit of the experimental data.

Several problems were however experienced in fitting the EOS to the experimental data at temperatures close to the CO_2 critical temperature. The problems with correlating the mixture critical region and the liquid compositions at these temperatures are highlighted in Fig. 9.

From the modelling results, it appears as if the Patel Teja equation of state is the most successful of the three equations in representing the experimental data. For the majority of the n-alkane systems, the PT provides the best fit of the data and gives consistently satisfactory results as indicated in Fig. 10. As can be seen in Fig. 11, the type III, phase behaviour as observed in the CO_2–n-C_{36} system, is not captured by any of the EOS using the interaction parameters that best fit the rest of the phase diagrams. As expected, the cubic equations of state also gave a poor representation of the system densities. This can be seen in Fig. 12.

From Figs. 13 and 14 it appears as if the PT EOS binary interaction parameters exhibit a functional relationship with system temperature and the n-alkane acentric factor. This makes it possible to estimate interaction parameters for other CO_2–n-alkane systems not studied in this work.

9. Conclusions and future work

The phase equilibria and densities of binary mixtures of supercritical CO_2 with n-C_{20}, n-C_{24}, n-C_{28} and n-C_{36} have been measured in the temperature range of 313–367 K. For the system CO_2–n-C_{36} the existence of another liquid–vapour dome at low n-alkane concentrations was found. This phenomenon warrants further investigation.

From the modelling results, it appears as if the Patel Teja equation of state gives a reasonably accurate representation of the experimental phase equilibria. All the EOS were unable to capture the additional vapour–liquid equilibrium, exhibited by the CO_2–n-C_{36} system, and also failed to model the binary mixture densities at the phase boundaries.

The Patel–Teja parameters exhibit a definite functional relationship with the n-alkane acentric factor and to a lesser degree with the system temperature. By extending the range of systems investigated to incorporate a broader range of n-alkanes and system temperatures it would be possible to develop a general interaction parameter correlation.

It is evident that more research is needed to improve the performance of equations of state in the mixture critical region.

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